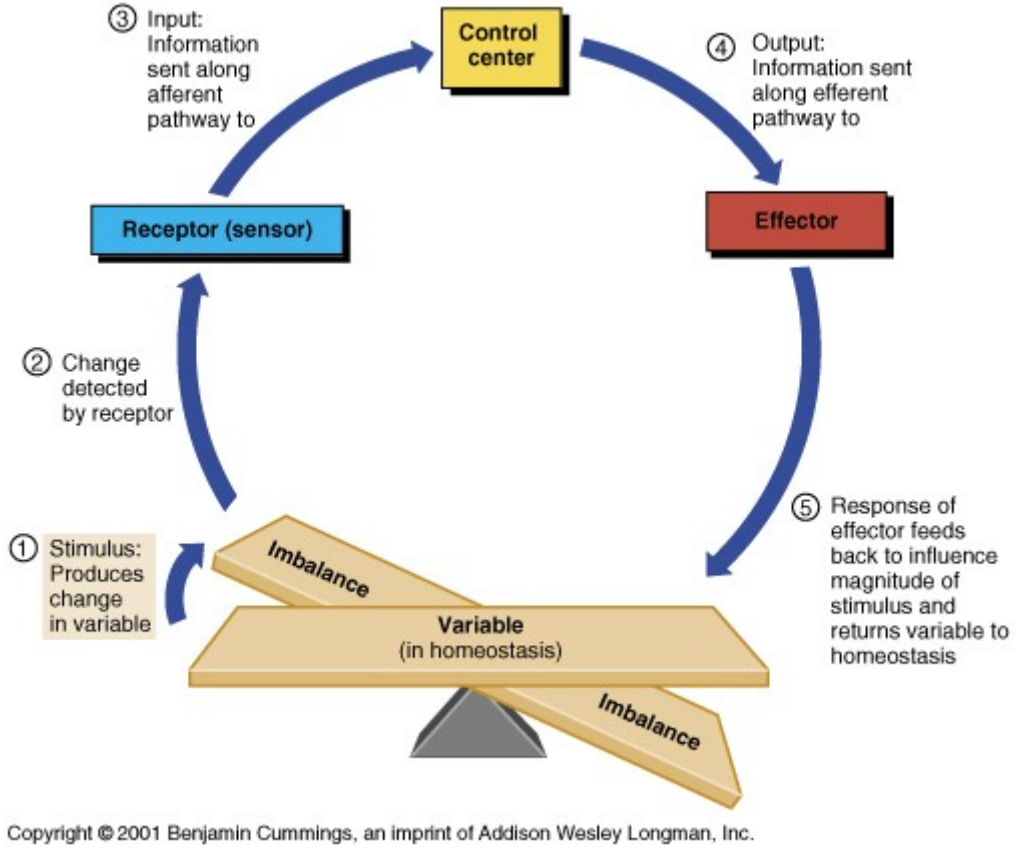
Homeostasis



Intro to Homeostasis

Homeostasis

- Maintenance of internal body conditions within a narrow range of values – a “steady state”, so that organs and systems can continue to function.

Eg: Body temperature of 37°C:

- Allows chemical reactions within cells to proceed at the correct pace
- “Not too hot, not too cold”
- Allows proteins to maintain correct shape

Eg: Neutral fluid pH

- Allows chemical reactions within cells to proceed at correct pace
- Helps with management of CO_2 levels

Eg: Steady supply of glucose for cellular respiration

Eg: Steady supply of oxygen for cellular respiration



Learning Aim: Define homeostasis and homeostatic mechanisms

Homeostatic mechanisms

- Are the mechanisms that maintain internal conditions
- Allow some independence from external environment and therefore a survival advantage

Eg: mechanisms like increased metabolic rate, shivering, and changes to blood flow allow us to maintain body temperature even when it is cold outside.

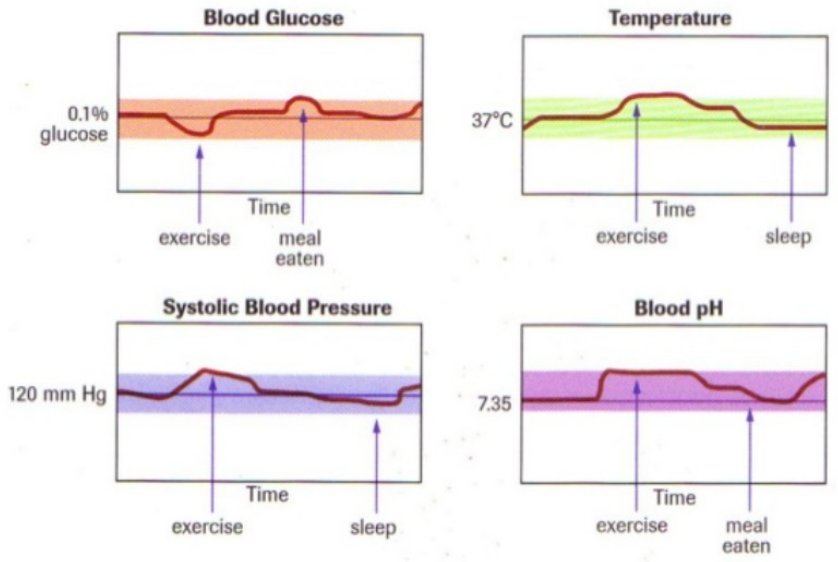


Learning Aim: Define homeostasis and homeostatic mechanisms

Steady State and Dynamic Equilibrium

Homeostasis maintains a "***steady state***" – a balanced internal environment in terms of:

- Core body temperature
- pH and concentration of other dissolved substances
- Glucose concentration
- O2 and CO2 concentration
- Blood pressure
- Metabolic waste concentration



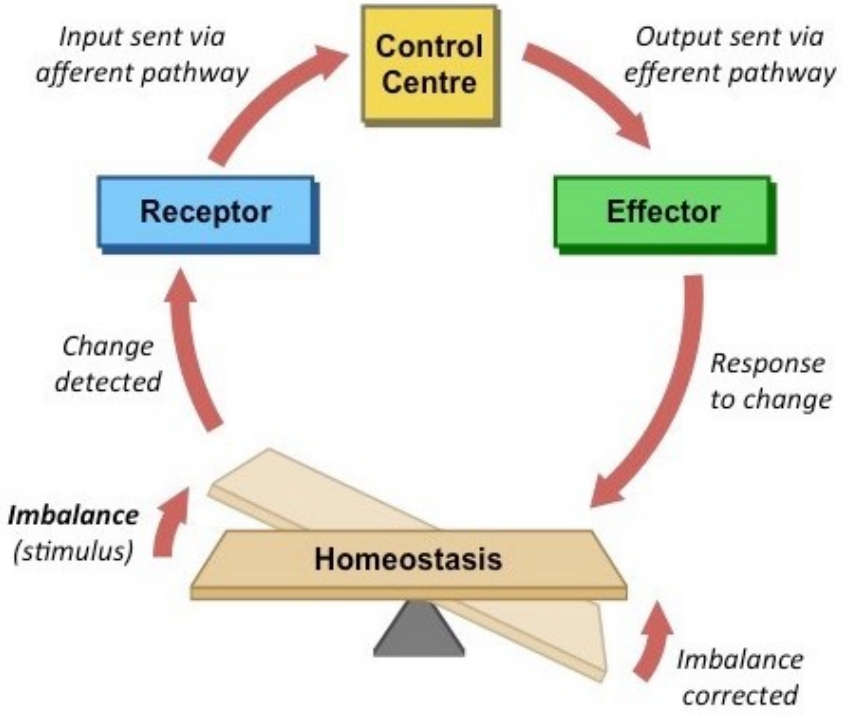
Dynamic equilibrium:

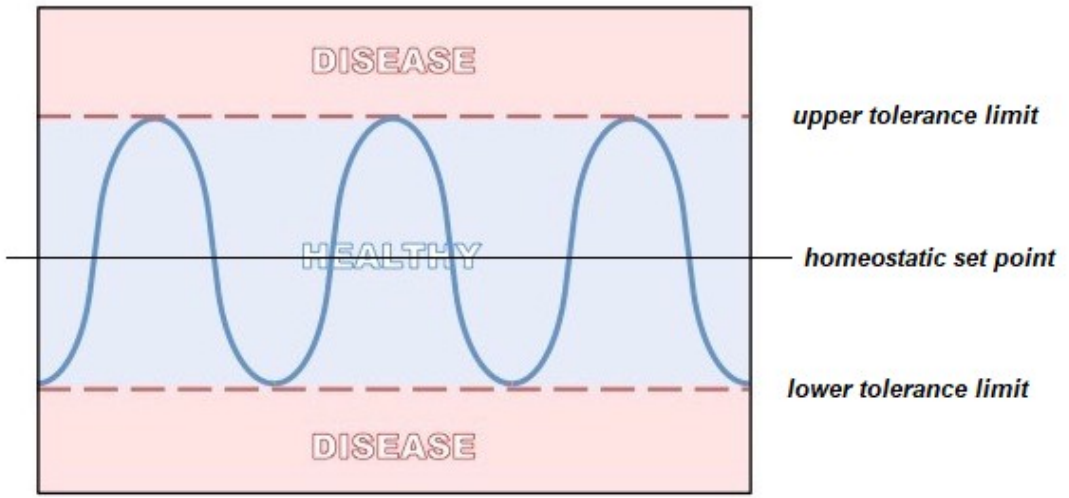
- The internal environment fluctuates within a narrow, acceptable range
- There is a constant process of sensing, responding and feedback to maintain this.

Learning Aim: Define steady state and dynamic equilibrium

Tolerance limits:

- The upper and lower limits of the factors in the body
- Normal function within tolerance limits
- Eg: the tolerance limits for blood glucose are 4.0-7.8mmol/L
- If tolerance limits are breached (too high or too low), dysfunction will occur.





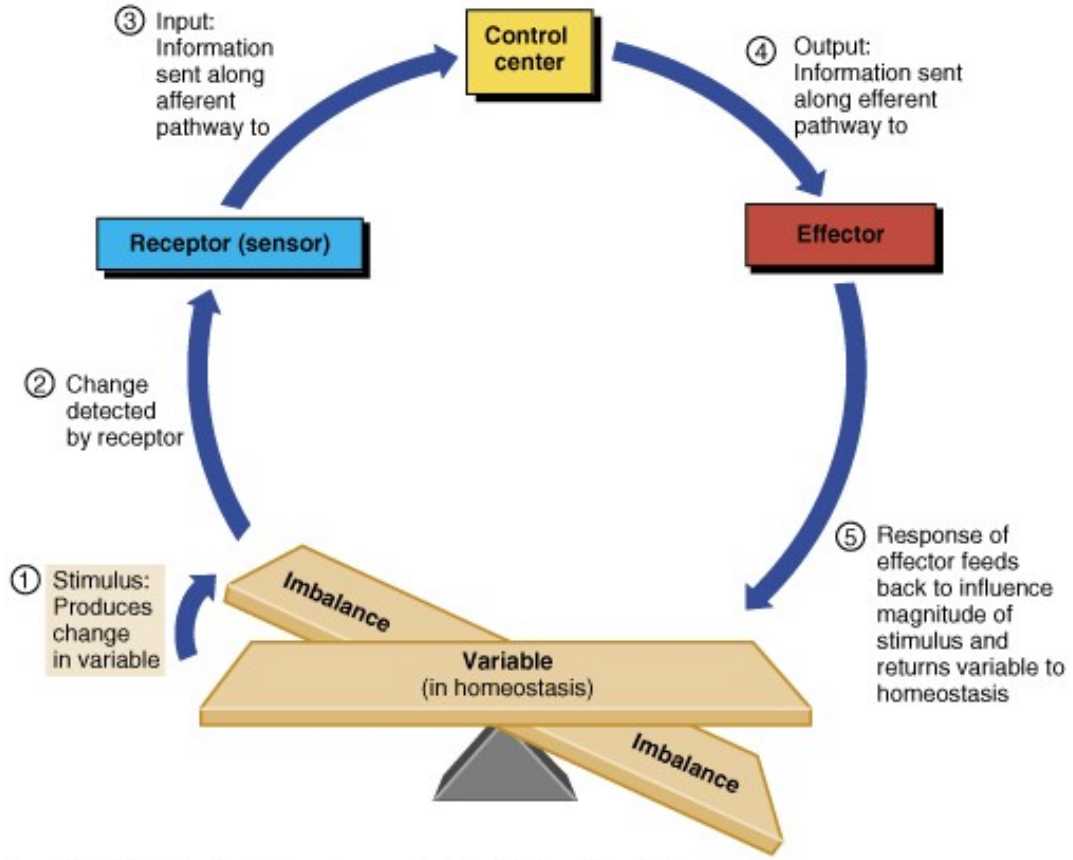
Homeostasis does **not** involve keeping conditions static
It involves keeping conditions within tightly regulated *physiological tolerance limits*

Learning Aim: Define homeostasis and homeostatic mechanisms

Homeostasis and Feedback Loops

Key Terms:

- *Stimulus:* a change in the environment that can be detected and acted on.
- *Receptor:* a part of the body that detects the change.
- *Message:* how information is sent through the body.
- *Modulator (control centre):* a part of the body that processes the information from the receptor, and sends information to the effector.
- *Effector:* a part of the body that carries out the response to the stimulus, counteracting it and returning balance.
- *Response:* how the effector responds to the message from the modulator
- *Feedback*: achieved when the original stimulus has been changed by the response.



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Learning Aim: List and describe the key components of a homeostatic feedback loop

A homeostatic feedback loop

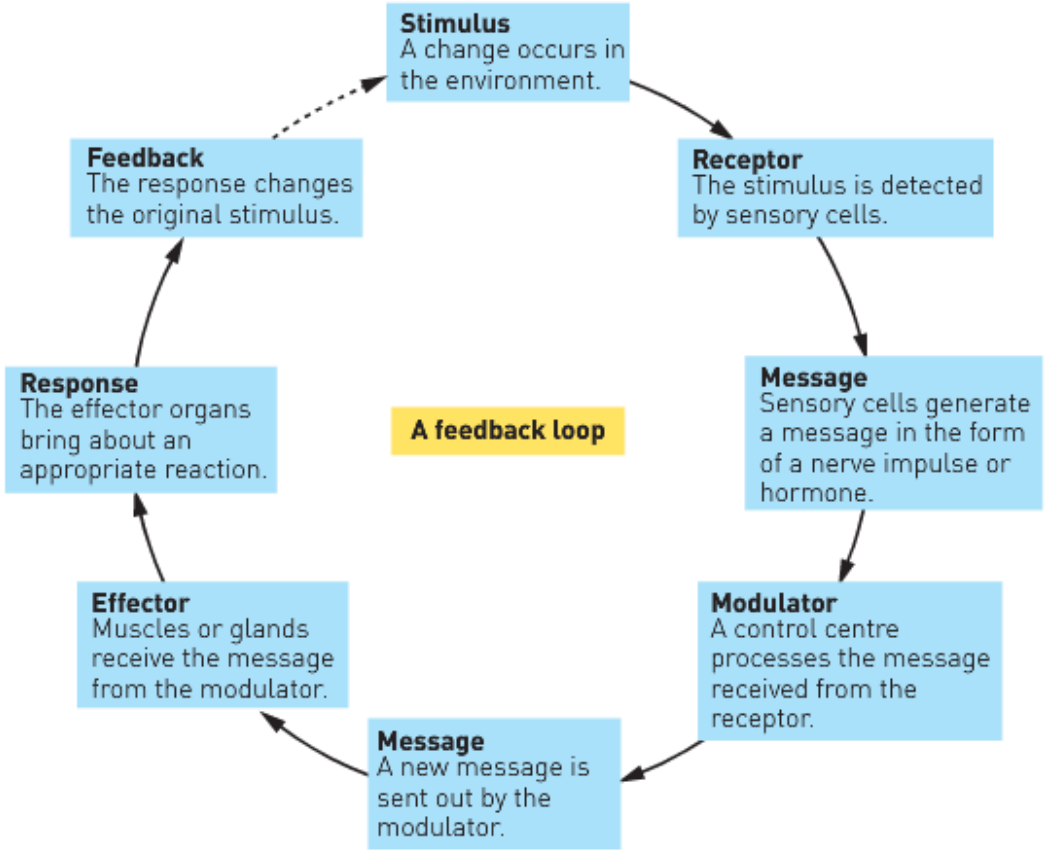
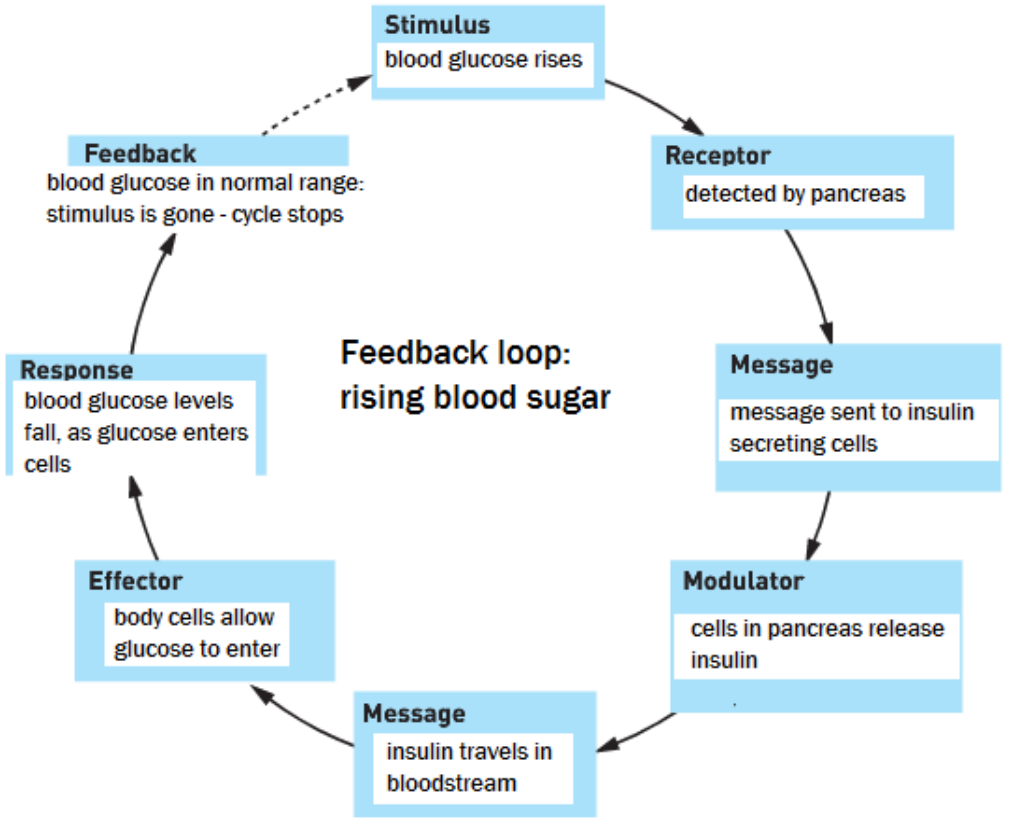


Figure 6.5 The stimulus-response-feedback model



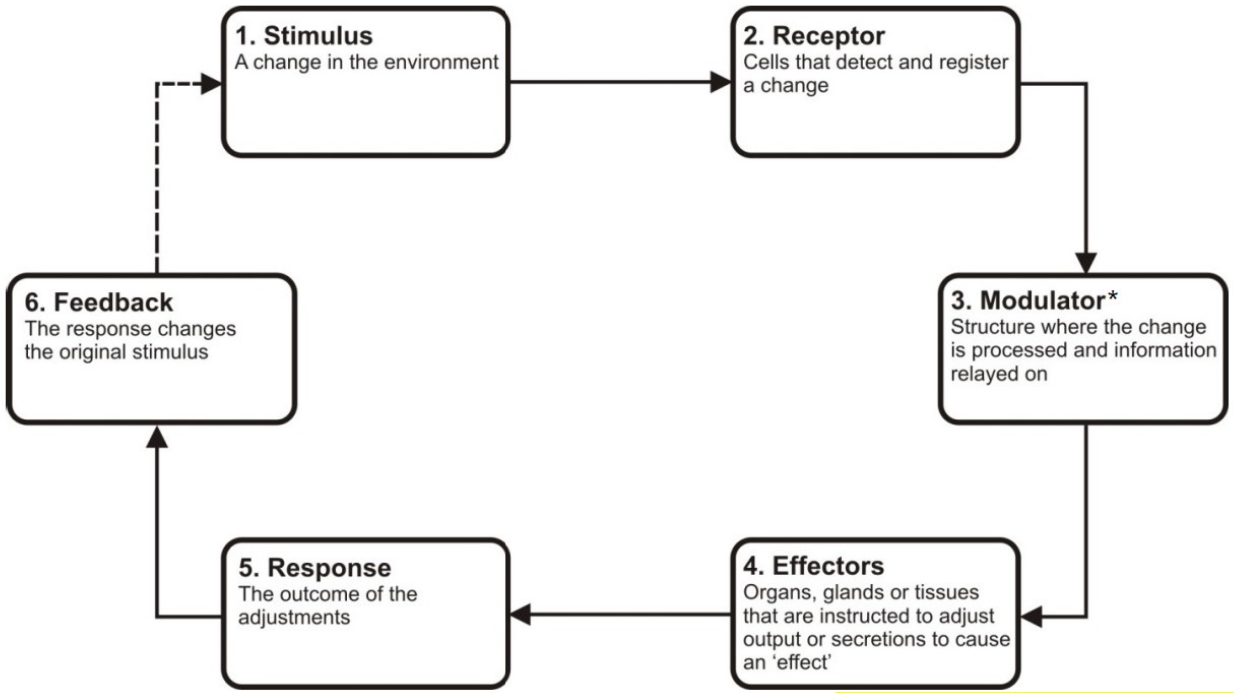
Learning Aim: List and describe the key components of a homeostatic feedback loop

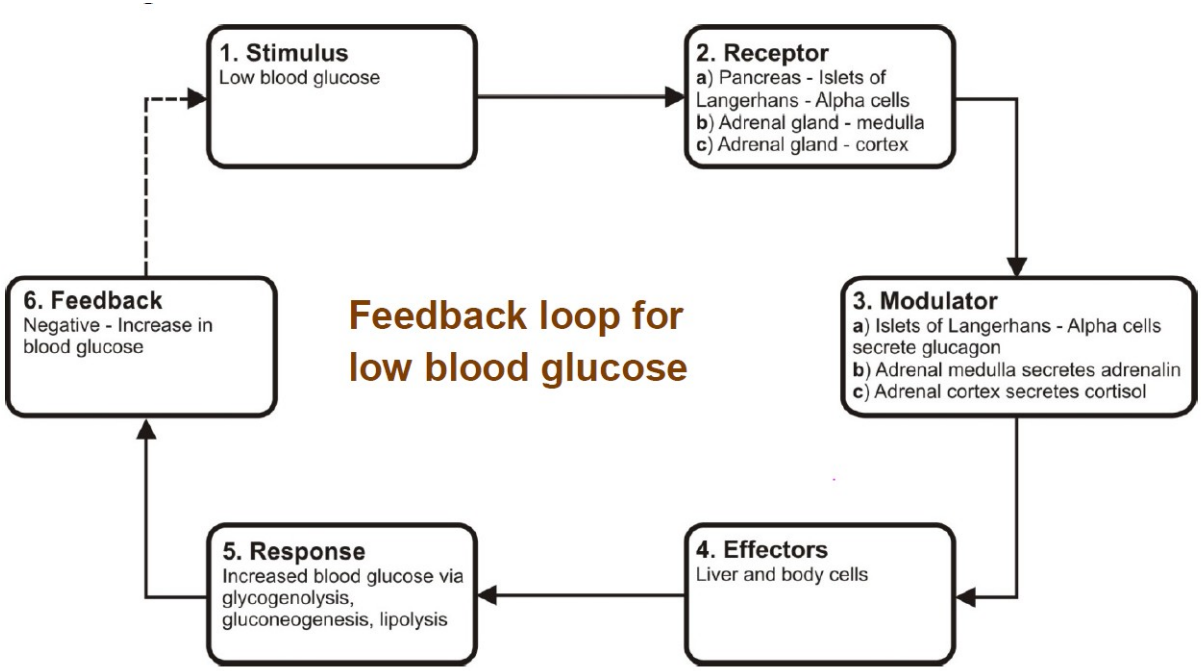
How to represent homeostatic loops in an exam question

A **stimulus-response feedback model (*steady state control model)*** is a good way of showing how mechanisms are operating in the body in order to maintain **homeostasis**.

Things to remember about this model:

- the loop represents a continuous process so there is no starting or finishing point
- the response alters the original stimulus



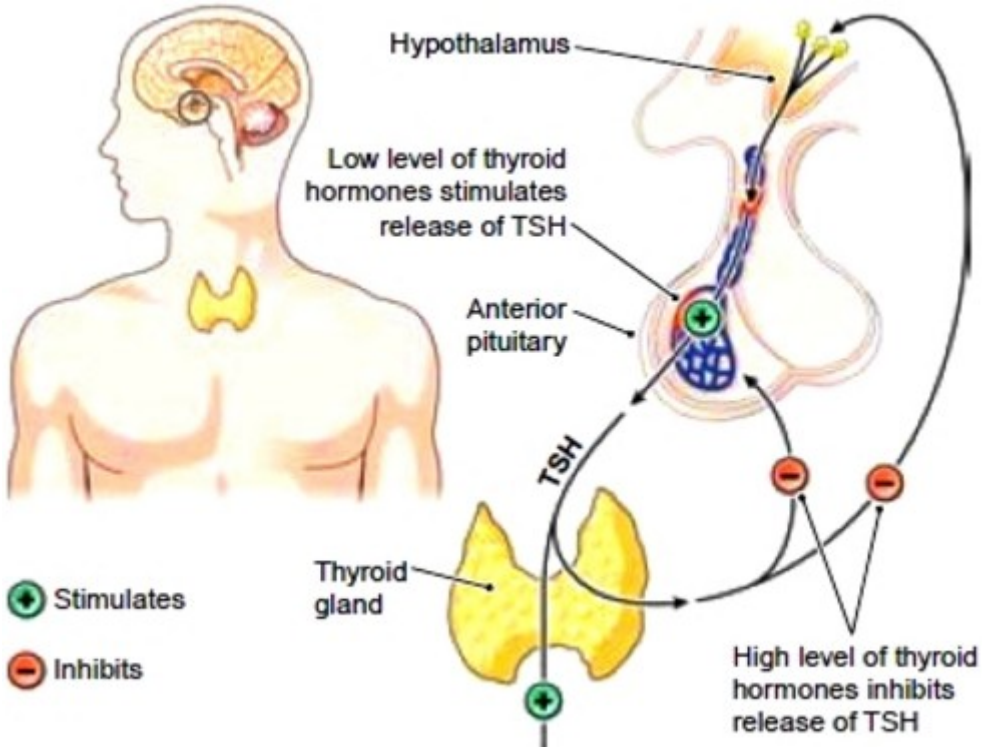


Learning Aim: List and describe the key components of a homeostatic feedback loop

Feedback Systems

Negative feedback:

- The response to the original stimulus reduces or eliminates the stimulus, and therefore the continued response.
- Eg: Thyroid hormone negative feedback



Learning Aim: Describe positive and negative feedback, giving examples

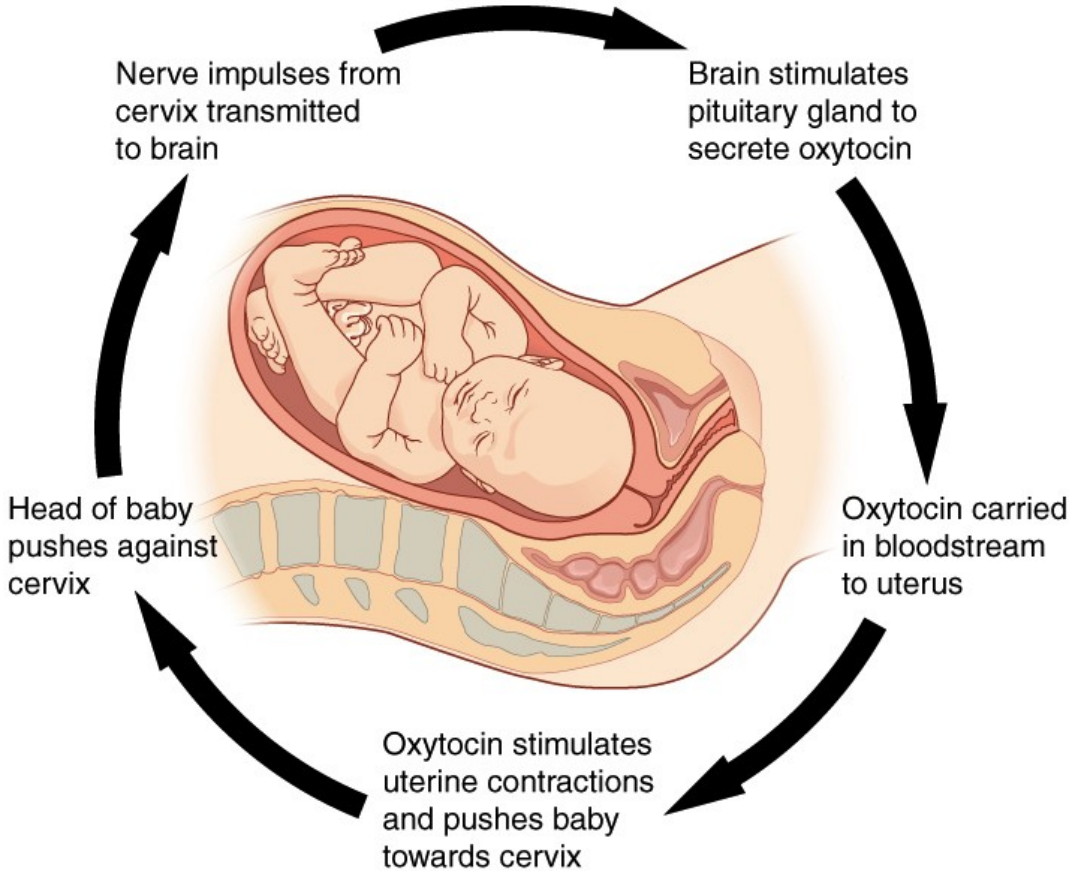
Feedback Systems

Positive feedback:

- Occurs in processes where stimulus needs to be reinforced for continual effect: not usually part of homeostasis.

- Examples: oxytocin release during labour

  blood clotting cascade



Learning Aim: Describe positive and negative feedback, giving examples

Body Temperature Homeostasis



Intro to thermoregulation

Thermoregulation when body temperature falls

Thermoregulation

- Regulation of core body temperature at 37°C set point
  - Temperature where chemical reactions in cells are stable and optimised.
- Several homeostatic mechanisms involved.
- Balance between heat production and heat loss.
- If body temp is too high:
  - Proteins denature
  - Nerve cells do not function correctly
  - Chemical reactions are uncontrolled
  - DEATH
- If body temp is too low:
  - Cellular reactions too slow
  - DEATH



Learning Aim: Define thermoregulation

Homeostatic mechanisms balance heat gain and heat loss to maintain an internal temperature around the set point of 37°C

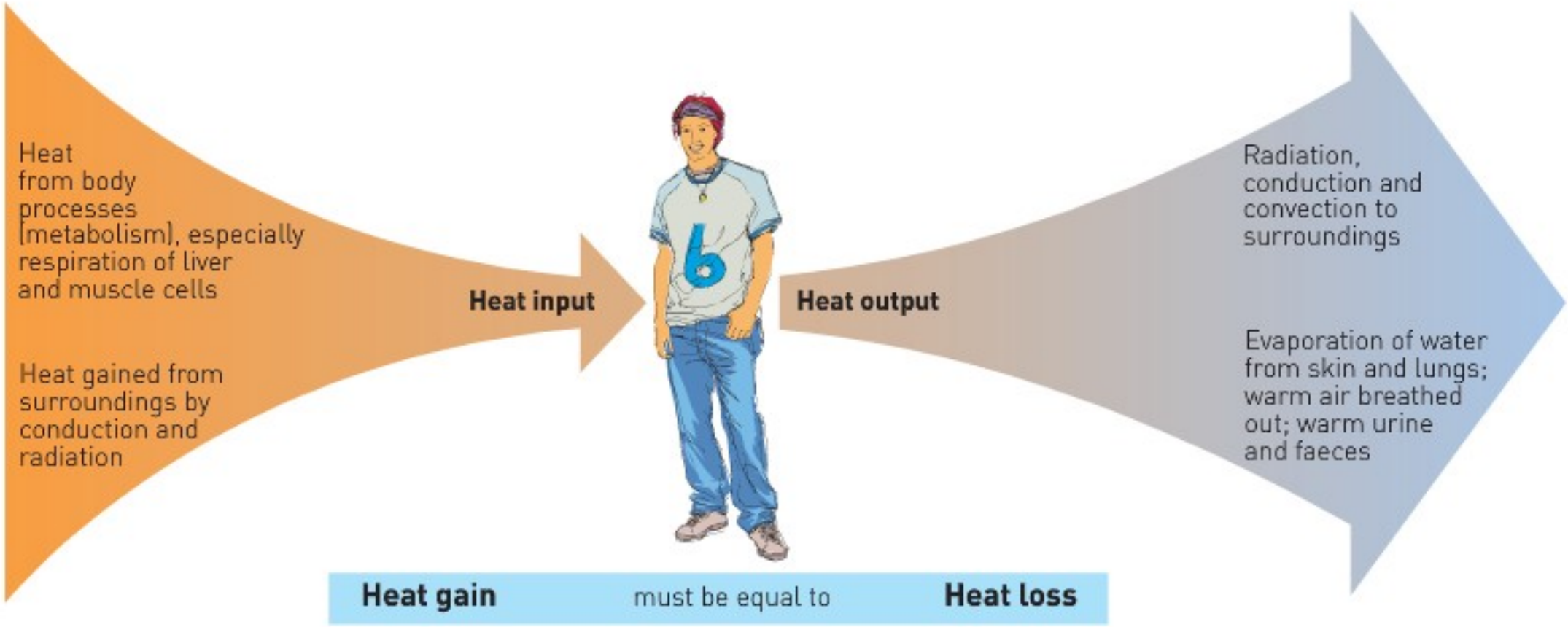


Figure 7.2 A constant human body temperature is achieved by a balance between heat gain and heat loss.

Learning Aim: Describe mechanisms of heat loss and gain

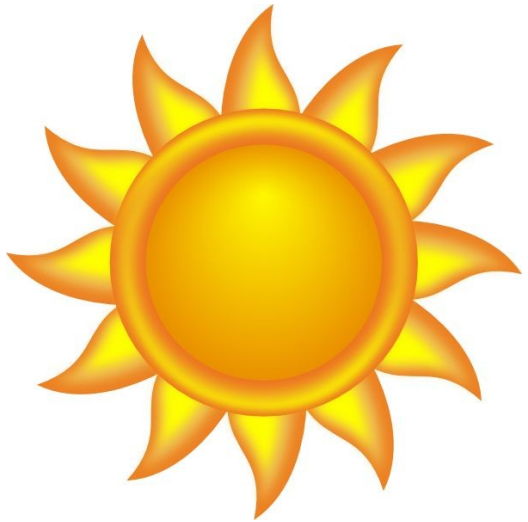
External Heat gain

- From external environment via conduction and radiation:

Conduction: warm air or water in contact with skin conducts heat into the body. Eg

- Warm air from the heater in a warm room.
- Warm water in a bath.
- Warm air outside on a hot day.

Radiation: solar radiation when standing in the sun. Radiation from a heater.



Learning Aim: Describe mechanisms of heat loss and gain

External Heat loss

- From body via convection

Convection: cold air or water in contact with skin moves heat away from the body. Eg

  - Cold air outside or in a cold room
  - Cold wind moves heat very quickly away from skin
  - Cold water in a bath or swimming pool
  - Sweat evaporates from skin,taking heat away from the skin



Learning Aim: Describe mechanisms of heat loss and gain

Internal Heat Gain

- During cellular respiration (metabolism)
- When metabolic rate increases, heat production increases.
  - Eg during Exercise:
    - Muscles need energy to contract
    - Cells respire more
    - More heat produced
    - Body heat rises
  - Things that can increase metabolic rate and body heat include:
    - Exercise
    - Stress
    - Sympathetic NS stimulation

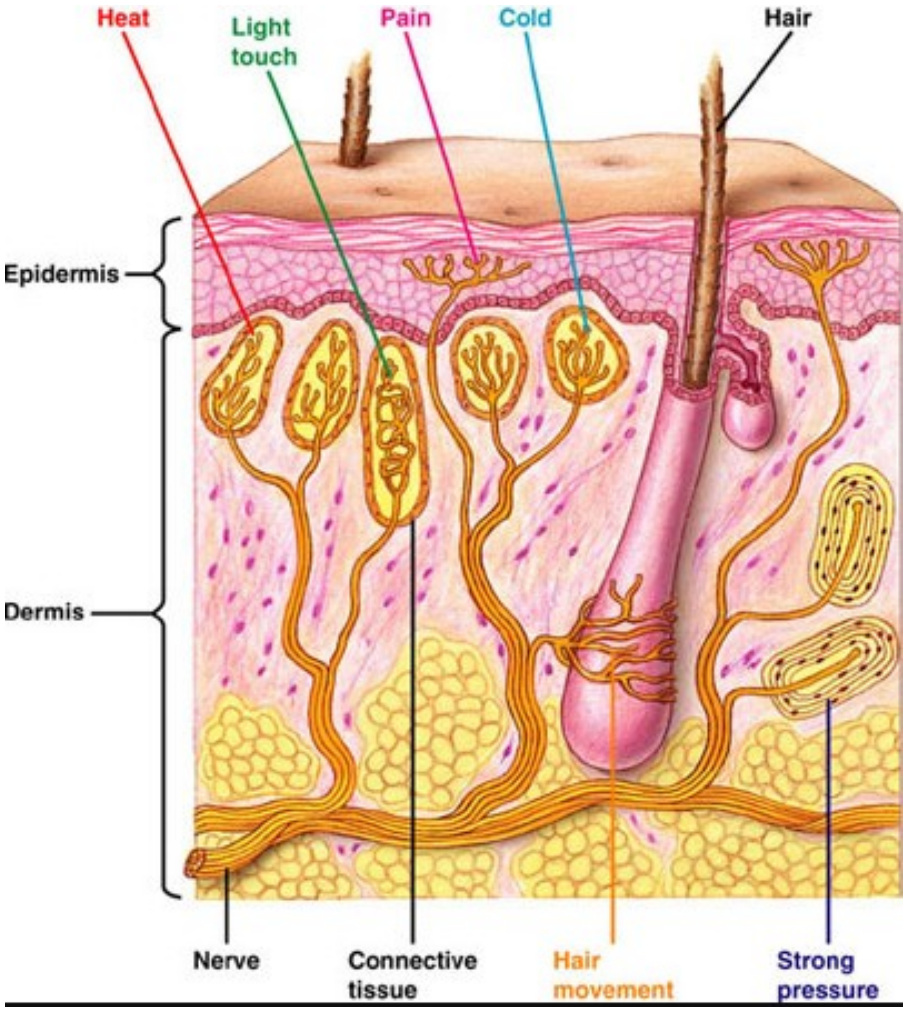


Learning Aim: Describe mechanisms of heat loss and gain

Temperature Receptors (thermoreceptors)

- Peripheral thermoreceptors:
  - In skin and other surface tissues
  - Detect temperature of external environment
    - Cold receptors
    - Heat receptors
  - Send signals to hypothalamus, which then acts to regulate body temperature

- Central thermoreceptors
  - In hypothalamus, spinal cord, abdominal organs
  - Send signals to hypothalamus, which then acts to regulate body temperature



Learning Aim: Identify locations and types of thermoreceptors involved in thermoregulation

Control Centres/Modulators for Thermoregulation

- Two main Control Centres/Modulators for Thermoregulation

1: The Hypothalamus

- Receives input from thermoreceptors in skin, and also its own thermoreceptors
- Send messages to effectors to regulate temperature

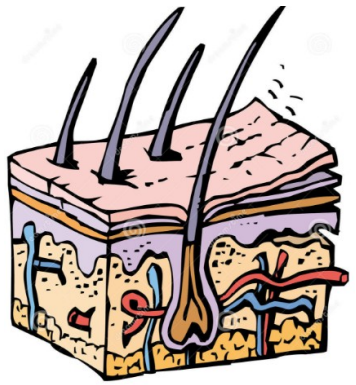
2: The Cerebral Cortex

- Receives input from thermoreceptors in skin and hypothalamus
- Integrates conscious awareness of body temperature
- Coordinates effectors to regulate temperature



Learning Aim: Identify modulators/control centres involved in thermoregulation

Effectors in Thermoregulation:



- Skin provides large surface area for heat exchange via
  - Conduction
  - Convection
  - Radiation
- Blood vessels help exchange heat between external and internal environments
  - Diameter of blood vessels can be controlled to increase or decrease this exchange
- Sweat glands in skin produce sweat when body is too hot
  - Evaporative cooling!

Note: evaporation from lungs also causes heat loss

- Muscle activity can be used to generate heat
- Endocrine glands can produce hormones that effect metabolism and therefore heat gain from cellular respiration

Learning Aim: Identify effectors involved in thermoregulation

Responses when body temperature falls below set point

- Vasoconstriction
  - Autonomic (sympathetic) stimulation from **hypothalamus** causes blood vessels to constrict.
  - Decreases flow of warm blood to skin, so less loss to cold environment via conduction, convection and radiation.
- Decreased Sweating
  - Autonomic stimulation from **hypothalamus** decreases sweat production, decreasing evaporative cooling

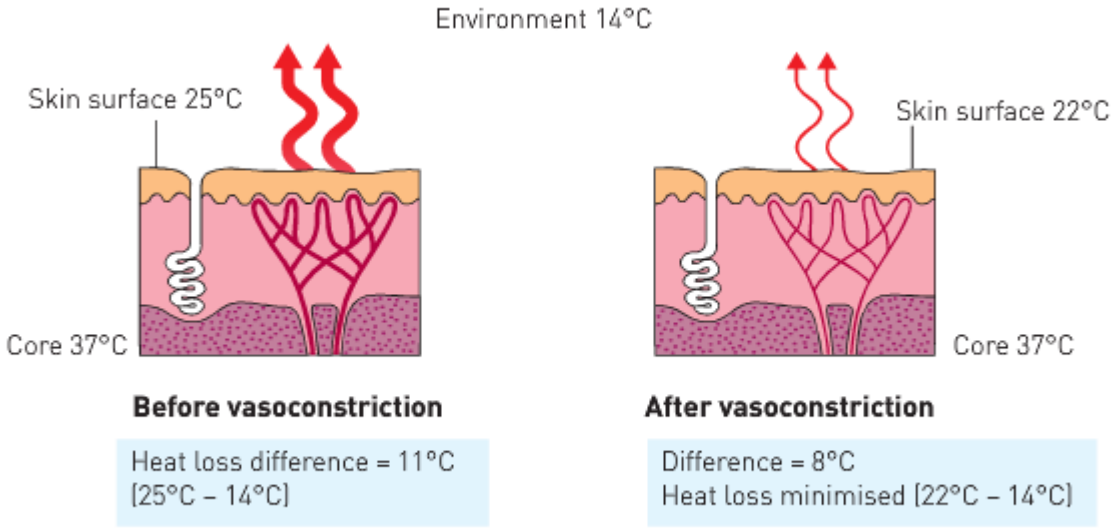


Figure 7.4 The effect of vasoconstriction in the skin

Learning Aim: Describe responses when body temperature falls below set point.

Responses when body temperature falls below set point

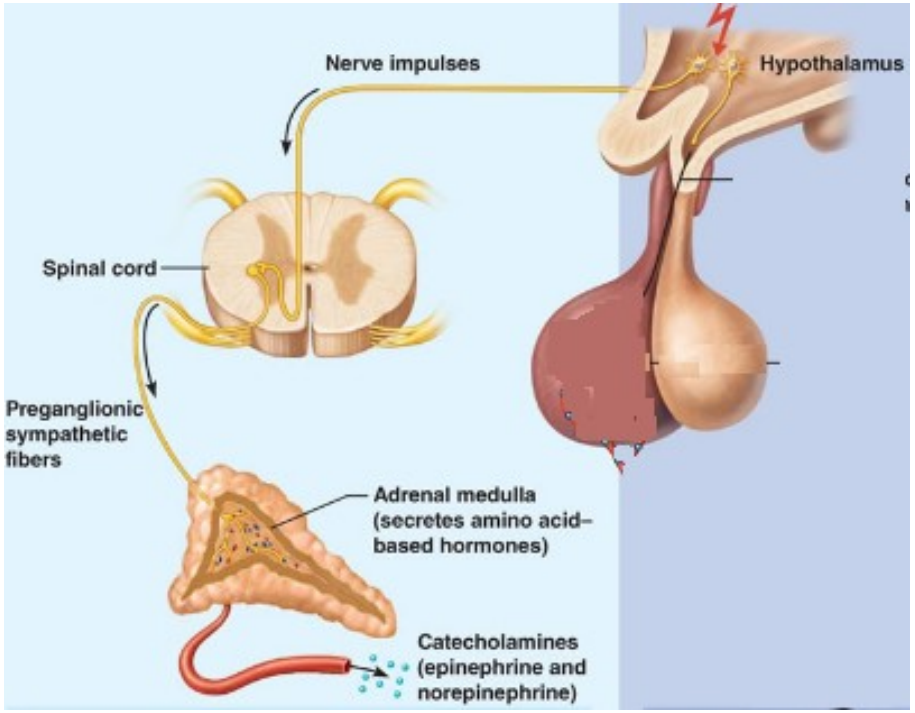
- Shivering
  - **Hypothalamus** sends message to brain centres that control skeletal muscle tone
  - Results in shivering – quick, repeated contraction of muscles, requiring cellular respiration.
  - Heat is released from muscles as a by-product of cellular respiration
  - Can be influenced by conscious thought from cerebral cortex (up to a point)



Learning Aim: Describe responses when body temperature falls below set point.

Responses when body temperature falls below set point

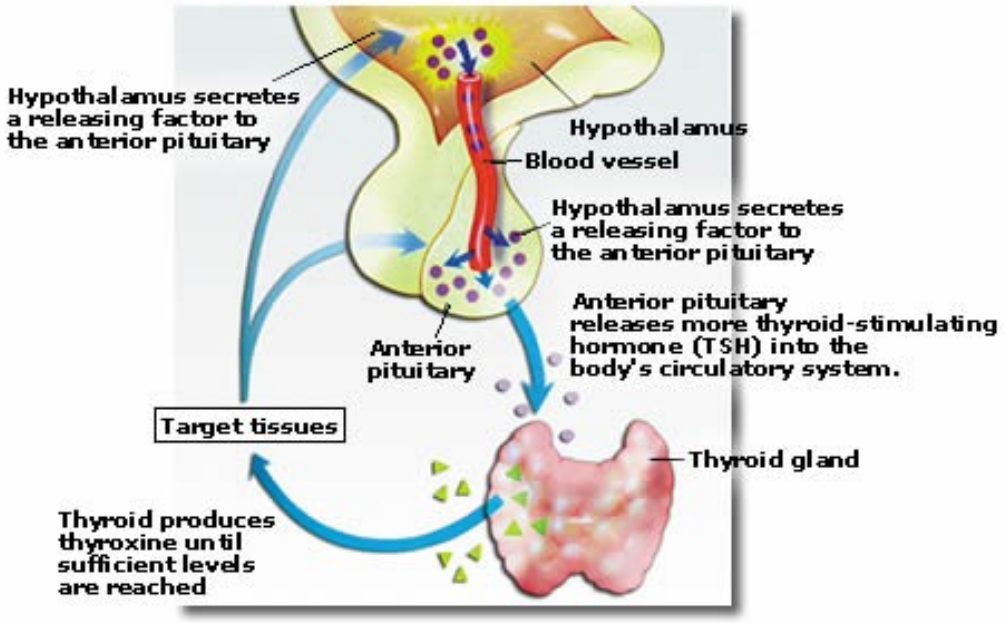
- Hypothalamic Sympathetic NS stimulation of adrenal medulla – endocrine response.
  - Adrenalin and Noradrenalin release.
  - Increase in cellular metabolism and heat production.
  - Counteracts heat loss.



Learning Aim: Describe responses when body temperature falls below set point.

Responses when body temperature falls below set point

- Increase in thyroxine production
  - Hypothalamus sends message to anterior pituitary for endocrine response
  - TSH (Thyroid Stimulating Hormone) released
  - Stimulates Thyroid
  - Thyroxine released
  - Increases metabolic rate and therefore heat.



Learning Aim: Describe responses when body temperature falls below set point.

Responses when body temperature falls below set point

Conscious and behavioural responses to heat loss:

- Conscious awareness of feeling cold leads to behavioural changes to minimise heat loss:
  - Seeking shelter from wind/rain/cold weather, and seeking warmth
  - Putting on clothing to insulate against heat loss
  - Curling up to decrease surface area for heat loss
  - Moving more to increase metabolic activity to increase heat gain



Learning Aim: Describe responses when body temperature falls below set point.

Homeostasis when body temperature falls



Stimulus

Falling Body Temperature

Receptor

Thermoreceptors in Skin (surface) and Hypothalamus, Spinal Cord and Abdominal Organs (core)

Message

Sensory Nervous System

Control Centre / Modulator

Hypothalamus

Cerebral Cortex Conscious Awareness)

Message

Sympathetic Nervous System

Neurons within brain

TrH to ant.pit, then TSH

Somatic Motor NS

Effector

Blood Vessels

Adrenal Medulla

Brain centres for muscle tone

Thyroid

Skeletal Muscles

Response

Vasoconstriction decreases blood flow to skin, reducing heat exchange

Adrenalin and Nor-adrenalin release leads to increased cellular metabolism and heat production to counter-act heat loss

Shivering leads to re-lease of heat from mus-cles to increase body temp

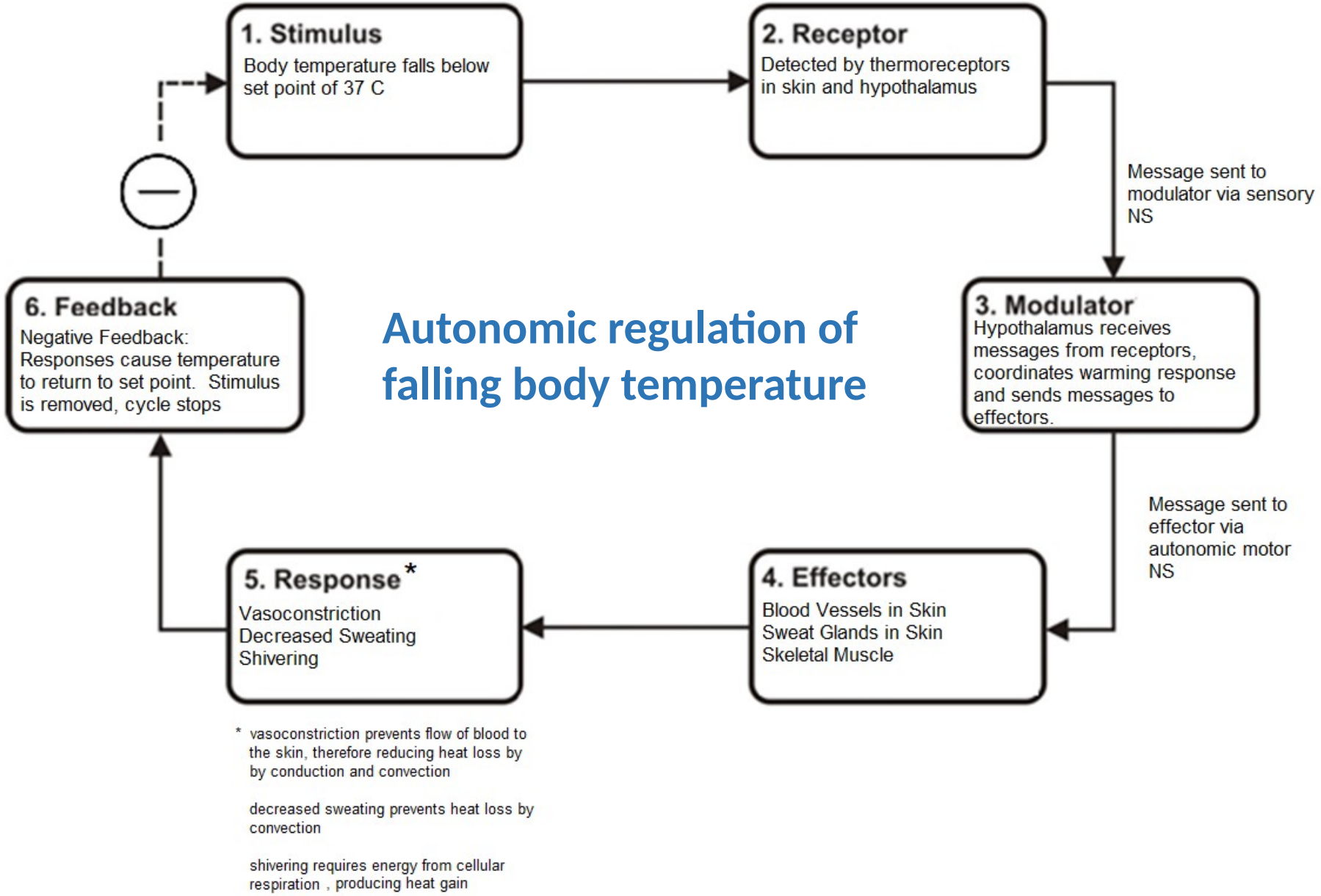
Produces thyroid hormones that increase metabolic rate and therefore heat produc-tion.

Behavioural responses such as seeking shelter, huddling, putting on clothes, turning on heater reduce heat loss.

Feedback

Negative Feedback: Body temperature returns to normal, stimulus no longer present, process ends.

Example of homeostatic feedback loop:



Note: you will need to be able to do feedback loops for all of the different ways body temperature is regulated – you can practice this on your review worksheet.

Learning Aim: Draw homeostatic feedback loops (steady state control models) to represent thermoregulation when body temperature falls.